Supplementary Material: Full Proofs for PERO

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Proofs

This supplement provides full proofs for the theoretical results in PERO. To make the presentation self-contained, we first restate the update rules, assumptions, and theorems used in the main paper, and then provide the detailed proofs.

Preliminaries

We first restate the update rules used in PERO. At iteration t, the classifier parameters are updated from $\boldsymbol{\theta}_{t-1}$ to $\boldsymbol{\theta}_t$ by performing a stochastic gradient descent step on the selected subset $S_t^B = \{(\mathbf{x}_t^{(i)}, y_t^{(i)})\}_{i=1}^B$:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \frac{\gamma_\theta}{B} \sum_{i=1}^{B} \nabla_{\boldsymbol{\theta}} \ell_t^{(i)}, \tag{1}$$

where γ_θ denotes the learning rate.

The pre-evaluation module is trained online to approximate the sample-wise classification loss. At iteration t, the classifier $f_{\boldsymbol{\theta}_{t-1}}$ computes the true loss for each sample in the selected subset:

$$\ell_t^{(i)} \coloneqq \ell(f_{\boldsymbol{\theta}_{t-1}}(\mathbf{x}_t^{(i)}), y_t^{(i)}), \quad i = 1, \cdots, B.$$

The module parameters $\boldsymbol{\phi}_t$ are then updated by minimizing the mean squared error between the predicted losses $g_t^{(i)} \coloneqq g_{\boldsymbol{\phi}_t}(\mathbf{x}_t^{(i)})$ and the corresponding true losses $\ell_t^{(i)}$:

$$\mathcal{L}_{\mathrm{MSE}}(\boldsymbol{\phi}_t) = \frac{1}{B} \sum_{i=1}^{B} \left[g_t^{(i)} - \ell_t^{(i)} \right]^2, \quad \boldsymbol{\phi}_{t+1} = \boldsymbol{\phi}_t - \gamma_\phi \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\mathrm{MSE}}(\boldsymbol{\phi}_t), \tag{2}$$

where γ_ϕ denotes the learning rate of the pre-evaluation module.

We next recall the assumptions used in the theoretical analysis.

Assumption 1 (Bounded Loss)**.** $\ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y)$ is uniformly bounded: $\ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y) \leq L_{\max}$ for all $(\mathbf{x}, y)$.

Assumption 2 (Bounded Spectral Radius)**.** $\ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y)$ is twice continuously differentiable with respect to $\boldsymbol{\theta}$, and its Hessian satisfies $\lambda_{\max}\big(\nabla^2_{\boldsymbol{\theta}\boldsymbol{\theta}} \ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y)\big) \leq 1$ for all $(\mathbf{x}, y)$.

Assumption 3 (Lipschitz Continuity)**.** $g_{\boldsymbol{\phi}}(\mathbf{x})$ is Lipschitz continuous with respect to $\boldsymbol{\phi}$, i.e., $|g_{\boldsymbol{\phi}_1}(\mathbf{x}) - g_{\boldsymbol{\phi}_2}(\mathbf{x})| \leq L \|\boldsymbol{\phi}_1 - \boldsymbol{\phi}_2\|_2$ for all $\mathbf{x}$ and some constant $L > 0$.

Under these assumptions, we restate the two theoretical results established for PERO. The first result characterizes convergence to stationary points, while the second provides a generalization bound for the pre-evaluation module.

Theorem 1 (Convergence to Stationary Points)**.** *Under Assumptions 1, 2 and 3, suppose that the prediction error holds: $g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) - \ell^{(i)} < \delta, \forall i$ for a constant $\delta > 0$, and that the spectral radius of $\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g(\mathbf{x})$ is bounded by $\frac{1}{\delta}\left(\frac{1}{2} - L^2\right)$. Then, the sequence $\{\boldsymbol{\theta}_t, \boldsymbol{\phi}_t\}_t$ converges to a stationary point $(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$.*

Theorem 2 (Generalization Bound for the Pre-Evaluation Module)**.** *Under Assumption 1, the following generalization bound holds with a probability of at least $1 - \epsilon$ for any $\epsilon \in (0, 1)$:*

$$\left| R(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) - \widehat{R}(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) \right| < \sqrt{\left(\mathbb{V}[\ell] + \frac{L_{\max}}{3} \right) \frac{2 \ln\left(\frac{1}{\epsilon}\right)}{\hat{B}}} + L_{\max} \sqrt{\frac{\hat{B}}{2B^2} \ln\left(\frac{1}{\epsilon}\right)} + \frac{L_{\max}(\hat{B} - B)}{\hat{B}},$$

with $\mathbb{V}[\cdot]$ denoting variance and $L_{\max}$ as in Assumption 1.

Proof of Theorem 1

Proof. We first analyze the convergence of $\{\boldsymbol{\theta}_t\}_t$. Define $\mathcal{F}(\boldsymbol{\theta}) \coloneqq \frac{1}{B} \sum_{i=1}^{B} \ell(f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), y^{(i)})$ and let the resulting stationary point be $\boldsymbol{\theta}^*$. According to Eq. (1), we have

$$\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^* = \boldsymbol{\theta}_t - \boldsymbol{\theta}^* - \gamma_\theta \nabla_{\boldsymbol{\theta}} \mathcal{F}(\boldsymbol{\theta}_t). \tag{3}$$

By adopting the first-order Taylor expansion of the $\boldsymbol{\theta}$ related function $\nabla_{\boldsymbol{\theta}} \mathcal{F}(\boldsymbol{\theta})$ around $\boldsymbol{\theta}^*$, we can have

$$\nabla_{\boldsymbol{\theta}} \mathcal{F}(\boldsymbol{\theta}_t) = \nabla_{\boldsymbol{\theta}} \mathcal{F}(\boldsymbol{\theta}^*) + \nabla^2_{\boldsymbol{\theta}\boldsymbol{\theta}} \mathcal{F}(\boldsymbol{\theta}^*)(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*) + o\big((\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)\big) \tag{4}$$

With Eq. (3), and (4), we can drive the equation that

$$\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\| \tag{5}$$

$$= \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^* - \nabla^2_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathcal{F}(\boldsymbol{\theta}^*)(\boldsymbol{\theta}_t - \boldsymbol{\theta}^*) - o\big((\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)\big)\| \tag{6}$$

$$\leq \|(\mathbf{I} - \nabla^2_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathcal{F}(\boldsymbol{\theta}^*)\|\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| + o\big((\boldsymbol{\theta}_t - \boldsymbol{\theta}^*)\big). \tag{7}$$

According to Assumption 2, we know that

$$\lim_{t\to\infty} \frac{\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}^*\|}{\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|} \leq \|\mathbf{I} - \nabla^2_{\boldsymbol{\theta}\boldsymbol{\theta}}\mathcal{F}(\boldsymbol{\theta}^*)\| < 1. \tag{8}$$

We then analyze the convergence of $\{\boldsymbol{\phi}_t\}_t$. Let the resulting stationary point be $\boldsymbol{\phi}^*$.

$$\nabla^2_{\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}) = \frac{2}{B}\sum_{i=1}^{B}\left[g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) - \ell^{(i)}\right] \cdot \nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) \tag{9}$$

$$\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}) = \frac{2}{B}\sum_{i=1}^{B}\left[\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})^\top + \left[g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) - \ell^{(i)}\right] \cdot \nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right]. \tag{10}$$

According to Eq. (2), we have

$$\boldsymbol{\phi}_{t+1} - \boldsymbol{\phi}^* = \boldsymbol{\phi}_t - \boldsymbol{\phi}^* - \gamma_\phi \nabla_{\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}_t). \tag{11}$$

Similarly, by adopting the first-order Taylor expansion of the $\boldsymbol{\theta}$ related function $\nabla_{\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi})$ around $\boldsymbol{\phi}^*$, we can have

$$\nabla_{\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}_t) = \nabla_{\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}^*) + \nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}^*)(\boldsymbol{\phi}_t - \boldsymbol{\phi}^*) + o\big((\boldsymbol{\phi}_t - \boldsymbol{\phi}^*)\big) \tag{12}$$

With Eq. (11), and (12), we can drive the equation that

$$\begin{aligned}
&\|\boldsymbol{\phi}_{t+1} - \boldsymbol{\phi}^*\| \\
=&\|\boldsymbol{\phi}_t - \boldsymbol{\phi}^* - \nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}^*)(\boldsymbol{\phi}_t - \boldsymbol{\phi}^*) - o\big((\boldsymbol{\phi}_t - \boldsymbol{\phi}^*)\big)\| \\
\leq&\|(\mathbf{I} - \nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}^*)\|\|\boldsymbol{\phi}_t - \boldsymbol{\phi}^*\| + o\big((\boldsymbol{\phi}_t - \boldsymbol{\phi}^*)\big).
\end{aligned}$$

Given that the spectral radius of $\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x})$ is bounded by $\frac{1}{\delta}\left(\frac{1}{2} - L^2\right)$, we can have

$$\begin{aligned}
&\lambda_{\max}(\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi})) \leq \left\|\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi})\right\|_2 \\
&= \frac{2}{B}\left\|\sum_{i=1}^{B}\left[\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})^T + \left(g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) - \ell^{(i)}\right)\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right\|_2\right] \\
&\leq \frac{2}{B}\sum_{i=1}^{B}\left\|\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})^T + \left(g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)}) - \ell^{(i)}\right)\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right\|_2 \\
&\leq \frac{2}{B}\sum_{i=1}^{B}\left[\left\|\nabla_{\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right\|_2^2 + \delta \cdot \left\|\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right\|_2\right] \\
&\leq \frac{2}{B}\sum_{i=1}^{B}\left(L^2 + \delta\left\|\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\right\|_2\right) = 2\left(L^2 + \delta\lambda_{\max}\big(\nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}} g_{\boldsymbol{\phi}}(\mathbf{x}^{(i)})\big)\right) < 1.
\end{aligned}$$

As a consequence, we have

$$\lim_{t\to\infty} \frac{\|\boldsymbol{\phi}_{t+1} - \boldsymbol{\phi}^*\|}{\|\boldsymbol{\phi}_t - \boldsymbol{\phi}^*\|} \leq \|\mathbf{I} - \nabla^2_{\boldsymbol{\phi}\boldsymbol{\phi}}\mathcal{L}_{\text{MSE}}(\boldsymbol{\phi}^*)\| < 1. \tag{13}$$

In summary, we complete the proof of Theorem 1. □

Proof of Theorem 2

Denote $R(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) = \mathbb{E}_{p(\mathbf{x},y)}\left[\ell\right]$, $\widehat{R}(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) = \frac{1}{\hat{B}}\sum_{i=1}^{\hat{B}} z^{(i)}\ell^{(i)}$, and $\widetilde{R}(\boldsymbol{\theta}^*) = \frac{1}{\hat{B}}\sum_{i=1}^{\hat{B}} \ell^{(i)}$.

Lemma 1. *Let $\ell^{(1)}, \ldots, \ell^{(\hat{B})}$ be i.i.d. random variables taking values in $[0, L_{\max}]$, and let $1 \leq k \leq \hat{B}$. Define*

$$\bar{\ell} = \frac{1}{\hat{B}}\sum_{i=1}^{\hat{B}} \ell^{(i)}, \qquad T_{\hat{B},k} = \frac{1}{k}\sum_{i=\hat{B}-k+1}^{\hat{B}} \ell_{\langle i \rangle},$$

where $\ell_{\langle 1\rangle} \leq \cdots \leq \ell_{\langle \hat{B}\rangle}$ are the order statistics. Set $\Delta_{\hat{B},k} = T_{\hat{B},k} - \overline{\ell}$. Then, for any $\epsilon \in (0,1)$, with probability at least $1-\epsilon$, we can have

$$|\Delta_{\hat{B},k}| \leq L_{\max}\left(\sqrt{\frac{\hat{B}}{2k^2}\ln\left(\frac{1}{\epsilon}\right)} + \frac{\hat{B}-B}{\hat{B}}\right).$$

Proof of Lemma 1. We first show that $\Delta_{\hat{B},k}$ satisfies the bounded-differences condition with constant $L_{\max}/k$. Define

$$\Delta_{\hat{B},k} = \frac{\hat{B}-k}{\hat{B}}\left(T_{\hat{B},k} - U_{\hat{B},k}\right),$$

where $U_{\hat{B},k} = \frac{1}{\hat{B}-k}\sum_{i=1}^{\hat{B}-k}\ell_{\langle i\rangle}$.

Let $\boldsymbol{\ell} = (\ell^{(1)},\ldots,\ell^{(\hat{B})})$ be the loss vector. Consider changing one coordinate $\ell^{(j)}$ to $\ell^{(j)\prime}$ to form $\boldsymbol{\ell}'$. The change in $T_{\hat{B},k}$ is at most $L_{\max}/k$, and the change in $U_{\hat{B},k}$ is at most $L_{\max}/(\hat{B}-k)$. Thus,

$$|(T_{\hat{B},k} - U_{\hat{B},k})(\boldsymbol{\ell}) - (T_{\hat{B},k} - U_{\hat{B},k})(\boldsymbol{\ell}')| \leq \frac{L_{\max}}{k} + \frac{L_{\max}}{\hat{B}-k}.$$

Multiplying by $(\hat{B}-k)/\hat{B}$ gives

$$|\Delta_{\hat{B},k}(\boldsymbol{\ell}) - \Delta_{\hat{B},k}(\boldsymbol{\ell}')| \leq \frac{\hat{B}-k}{\hat{B}}\left(\frac{L_{\max}}{k} + \frac{L_{\max}}{\hat{B}-k}\right) \tag{14}$$

$$= \frac{L_{\max}}{\hat{B}}\left(\frac{\hat{B}-k}{k} + 1\right) \tag{15}$$

$$= \frac{L_{\max}}{k}. \tag{16}$$

Hence $\Delta_{\hat{B},k}$ is a function of $\hat{B}$ independent random variables with bounded differences constant $L_{\max}/k$. Applying McDiarmid's inequality to $\Delta_{\hat{B},k}$ yields for any $\xi > 0$,

$$\mathbb{P}\Big(\Delta_{\hat{B},k} - \mathbb{E}[\Delta_{\hat{B},k}] \geq \xi\Big) \leq \exp\left(-\frac{2\xi^2}{\hat{B}L^2}\right) = \exp\left(-\frac{2k^2\xi^2}{\hat{B}L_{\max}^2}\right).$$

Since $\Delta_{\hat{B},k} \geq 0$ and $\mathbb{E}[\Delta_{\hat{B},k}] \geq 0$, we have for any $\xi > 0$,

$$\mathbb{P}\Big(\Delta_{\hat{B},k} \geq \mathbb{E}[\Delta_{\hat{B},k}] + \xi\Big) \leq \exp\left(-\frac{2k^2\xi^2}{\hat{B}L_{\max}^2}\right).$$

Setting ϵ to match the upper bound, we obtain that, with probability at least $1-\epsilon$,

$$|\Delta_{\hat{B},k}| \leq L_{\max}\left(\sqrt{\frac{\hat{B}}{2k^2}\ln\left(\frac{1}{\epsilon}\right)} + \mathbb{E}[\Delta_{\hat{B},k}]\right). \tag{17}$$

Finally, since

$$\mathbb{E}[\Delta_{\hat{B},k}] \leq \frac{L_{\max}(\hat{B}-B)}{\hat{B}}, \tag{18}$$

combining the inequalities (17) and (18), we can obtain the result in Lemma 1. □

Proof of Theorem 2. Notice that $R(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*) - \widehat{R}(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*)$ can be decomposed to two parts, i.e.,

$$R(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*) - \widehat{R}(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*) = \Big(R(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*) - \widetilde{R}(\boldsymbol{\theta}^*)\Big) + \Big(\widetilde{R}(\boldsymbol{\theta}^*) - \widehat{R}(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*)\Big). \tag{19}$$

For the first part (i.e., $R(\boldsymbol{\theta}^*,\boldsymbol{\phi}^*) - \widetilde{R}(\boldsymbol{\theta}^*)$), we will adopt the Bernstein's inequality to provide an upper bound.

According to Assumption 1, we know that

$$\ell^{(i)} \leq L_{\max}, \forall i = 1,\cdots\hat{B}. \tag{20}$$

Thus, following Bernstein's inequality, we know that

$$\mathbb{P}\left(\left|\frac{1}{\hat{B}}\sum_{i=1}^{\hat{B}}\ell^{(i)} - R(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)\right| \geq \xi\right) \tag{21a}$$

$$= \mathbb{P}\left(\left|\widetilde{R}(\boldsymbol{\theta}^*) - R(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)\right| \geq \xi\right) \tag{21b}$$

$$\leq \exp\left(-\frac{\hat{B}\xi^2}{\frac{2}{\hat{B}}\sum_{i=1}^{\hat{B}}\mathbb{V}\left[\ell^{(i)}\right] + \frac{2\xi}{3}L_{\max}}\right). \tag{21c}$$

Setting ϵ to match the upper bound in inequality (21c) shows that with probability at least $1-\epsilon$, the following bound holds:

$$\left|R(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) - \widetilde{R}(\boldsymbol{\theta}^*)\right| \leq \sqrt{\frac{2\mathbb{V}[\ell]\ln\left(\frac{1}{\epsilon}\right)}{\hat{B}} + \frac{2L_{\max}\ln\left(\frac{1}{\epsilon}\right)}{3\hat{B}}}. \tag{22}$$

For the second part (i.e., $\widetilde{R}(\boldsymbol{\theta}^*) - \widehat{R}(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$), notice that $\widehat{R}(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$ can be regarded as the expectation of the largest B losses, therefore by applying Lemma 1, we can have that with probability at least $1-\epsilon$,

$$\left|\widetilde{R}(\boldsymbol{\theta}^*) - \widehat{R}(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)\right| \leq L_{\max}\sqrt{\frac{\hat{B}}{2B^2}\ln\left(\frac{1}{\epsilon}\right)} + \frac{L_{\max}(\hat{B}-B)}{\hat{B}}.$$

In summary, we complete the proof of Theorem 2. □